Astronomical
Satellite-based telescopes are ideal for observing space because, unlike terrestrial telescopes, they are not obstructed by Earth's atmosphere.

Telephones
Satellite phones exchange signals with satellites instead of ground-based cell towers. They are often used in remote areas not covered by terrestrial signals.

Defense
Military satellites have a range of uses, including surveillance, navigation, and for sending encrypted communications.

Television
Many television broadcasters deliver programs via satellite. Viewers receive the signals using a satellite dish attached to the outside of their home.

Uses of satellites

Although the first satellites were launched during the Cold War for the purposes of space exploration and defense, they are now engineered for a wide range of military and civilian uses. Most people make use of satellites every day without even realizing it.

Weather
Some satellites are designed to monitor features of Earth's weather and climate. They transmit the data back to Earth for analysis.

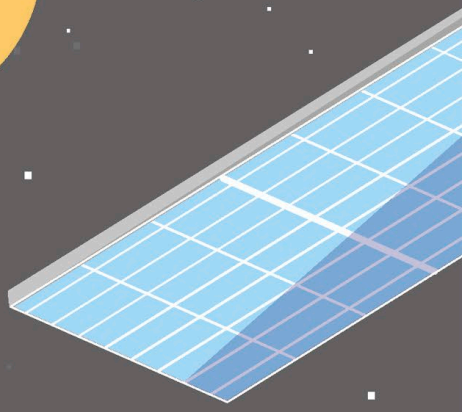
Radio
Relaying radio programs via a satellite means that signals can be transmitted across entire countries.

GPS navigation
Navigation devices can determine their position on Earth by exchanging information with satellites (see pp.194–195).

Internet
Satellite Internet provides coverage in remote areas, but service can be slow due to the distance signals have to travel.

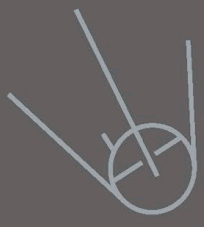
Satellites

Artificial satellites are specialized man-made spacecraft launched into orbit around Earth and other planets in the solar system. They are vital in telecommunications because they can receive signals from the ground and amplify and retransmit them to distant parts of Earth.



Communications satellites

Communications satellites are designed to send and receive radio signals carrying audio, video, and other kinds of data. Relaying signals via satellite allows for rapid communication across huge distances. Signals of set frequencies are sent into space from ground stations and picked up by a satellite's antenna. A transponder processes the information and boosts the signal before relaying it down to other ground stations on Earth.



SPUTNIK 1 WAS THE FIRST SATELLITE IN SPACE, LAUNCHED BY **THE SOVIET UNION ON OCTOBER 4, 1957**